Per-window Editing in GTA: Support-based μ and α

H2 study on halfcheetah-medium-v2, the reconstruction-score support, and follow-up tests H3–H7

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Summary

GTA edits every trajectory window with one global noising ratio μ and one return multiplier α. We asked whether these should be chosen *per window*.

test	result	verdict
H2a: more support $\Rightarrow$ higher safe μ	μ^{safe} rises $+0.18$ $[0.14, 0.23]$ from lowest to highest support	**supported**
H2b: higher quality $\Rightarrow$ lower μ	$\beta_Q \approx 0$; quality only predicts reward gain	**not supported**
H2c: support-adaptive μ (mean 0.3)	$+3.4$ over uniform 0.3, $+3.1$ over inverted (JAX)	**supported**
H2c-wide: adaptive μ (mean 0.5)	-0.7 vs uniform 0.5 ($p = 0.20$)	**not supported**
α by support	α_{up} $+0.9$ over $\alpha = 1.3$, but $=$ random mix ($p = 0.35$)	not specific
Learned $\mu(S,Q), \alpha(S,Q)$ (CMA-ES)	58.76 vs 59.05 for tuned global ($p = 0.29$)	**not supported**

Bottom line: support reliably predicts *how much physics damage* an edit does; turning that into a better RL score only helps when μ is small. At GTA's own operating point ($\mu = 0.5$, $\alpha = 1.3$) a single well-tuned global setting is as good.

Outline

1. Setup: GTA editing, μ and α
2. H2 made measurable: quality Q and three support metrics, including the **reconstruction score** S_2
3. Window-level study: pilot (S_1) $\rightarrow$ main (S_2)
4. Assigning μ by support: Experiment A (PyTorch and JAX TD3+BC), narrow and wide mappings
5. Assigning α by support: global sweep, up/down, random-mix control
6. Learning continuous $\mu(S, Q)$ and $\alpha(S, Q)$ with CMA-ES
7. Follow-up tests H3–H7 (no-generation baselines, VAE editors, trust weighting, bound)
8. Limitations and next steps

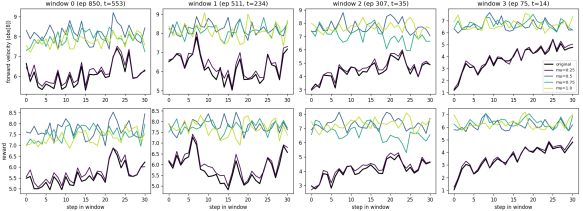
GTA in one slide (Lee et al., NeurIPS 2024)

1. **Train:** a conditional diffusion model (MixerUnet, 29.6M params, EDM, 1M steps) learns 32×24 trajectory windows given their return.
2. **Edit:** take a real window τ^0, noise it once to level $\sigma(\mu K)$, then denoise for μK steps conditioned on the *amplified* return $\alpha \cdot \text{RTG}$.
3. **Train RL:** add edited windows (as transitions) to the 1M original transitions; train TD3+BC.

$\mu \in (0, 1]$	how much of the window is rewritten (exploration). Paper: $\mu = 0.5$
$\alpha > 1$	how hard the edit is pushed to higher reward (exploitation). Paper: $\alpha = 1.3$
K	128 noise levels; μ is very non-linear in noise: $\sigma = 0.16$ at $\mu = 0.25$, 2.43 at $\mu = 0.5$

Data: D4RL `halfcheetah-medium-v2`, 1,000 episodes $\times$ 1,000 steps. Guidance $w = 0$ (config temperature 1).

Sanity check: GTA edits on random windows



Top: forward velocity; bottom: reward over 31 steps. Black = original.

- $\mu = 0.25$: almost no change ($+3\%$ reward).
- $\mu \geq 0.5$: $+35$–75% reward while staying physically valid in MuJoCo.
- Model-written reward matches the simulator within 0.5.

Every edit is replayed in MuJoCo to get the true reward (`oracle_r`), physics error (`dyn_mse`, paper Eq. 11) and distance from source (`shift`).

The hypothesis

H2. The best μ for a window depends on two of its properties:

- **H2a (support):** better covered by the data $\Rightarrow$ **higher** μ is safe.
- **H2b (quality):** higher own reward $\Rightarrow$ **lower** μ.
- **H2c (downstream):** choosing μ per window from support improves TD3+BC.

Quality: $q_i = \sum_{t=0}^{30} 0.99^t r_t$ inside the window; Q_i = percentile rank.

Support: three implementations, from data-only to model-based to exact:

	what it measures	cost	role
S_1	kNN distance in (s, a) space	cheap	pilot
S_2	diffusion **reconstruction score**	12 network calls	**primary**
S_3	exact log-likelihood (prob.-flow ODE)	very slow	validation

Support S_1: nearest neighbours in the data

1. Standardise every $(s, a) \in \mathbb{R}^{23}$ over all 1M transitions.
2. For each step t: distance to the **20th nearest** transition from a **different episode**

$$d_{20}(j_t) = \text{20th smallest of } \{\|z_{j_t} - z_{j'}\|_2 : \ \text{ep}(j') \neq \text{ep}(j_t)\}.$$

3. $\text{s1_dist}_i = \text{median}_t\, d_{20}(j_t)$; $S_{1,i} =$ percentile rank of $-\text{s1_dist}_i$.

- kNN density: $\hat{p}(x) \propto 1/d_k(x)^{23}$ — smaller distance, denser region. Inverse of GTA's *novelty* metric.
- Same-episode neighbours are excluded (adjacent steps are near-identical).
- Limitation: ignores the trained model; says nothing about what the *diffusion model* finds familiar.

Support S_2: the reconstruction score

1. Normalise the window $\tilde{x}$. For $\sigma \in \{0.16, 0.5, 2.4\}$ and 4 noise draws, one *unconditional* denoiser call:

$$\hat{x} = D_\theta(\tilde{x} + \sigma\varepsilon_n;\ \sigma, \varnothing)$$

2. Masked reconstruction error $e_i(\sigma) = \frac{1}{4}\sum_n \|M \odot (\hat{x} - \tilde{x})\|^2 / \sum M$.
3. s2_score$_i = -\operatorname{mean}_\sigma\ z\text{-score}_{\text{pool}}\big(\ln e_i(\sigma)\big)$; $\quad S_{2,i}$ = percentile rank.

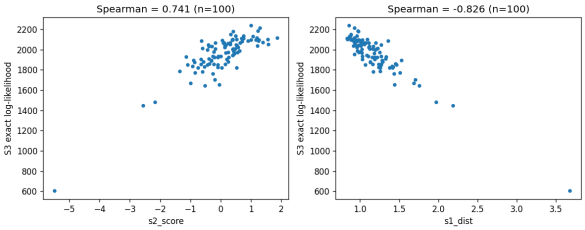
Why it measures support: $-\log p(x) \le \int w(\sigma)\, \mathbb{E}_\varepsilon \|D_\theta(x + \sigma\varepsilon; \sigma) - x\|^2 d\sigma + C$ (Kingma et al. 2021; Song et al. 2021). S_2 is a 3-point estimate of this bound.

- The three σ are exactly the noise levels an edit uses at $\mu \approx 0.25, 0.35, 0.5$: it asks *"can the model rebuild this window from the noise it will actually get?"*
- Cheap enough for all 161k windows used in RL experiments.

Support S_3 and agreement of the three metrics

S_3: exact $\log p_0(x)$ via the probability-flow ODE with Hutchinson divergence (dopri5); 100 windows only.

pair	Spearman ρ
S_1 vs S_2 (20k)	0.79
S_2 vs S_3 (100)	0.74
S_1 vs S_3 (100)	0.83
Q vs S_1	0.49
Q vs S_2	0.42



- Data-based and model-based support agree and both track the exact likelihood.
- Support and quality are related but separable ⇒ stratified 5×5 design on $Q \times S$.

From edits to "the best μ" of a window

For window i, with gain and physics error on the μ grid and threshold $\theta = 0.0085$ (paper's mean dyn. MSE):

- $\mu_i^* = \arg\max\{\text{gain}_i(\mu) : \ \text{dyn}_i(\mu) \le \theta\}$ — best reward among valid edits
- $\mu_{90,i} = \min\{\mu : \ \text{gain}_i(\mu) \ge 0.9 \max \text{gain}_i\}$ — how much editing it needs
- $\mu_i^{\text{safe}} =$ largest μ before dyn_i first exceeds θ — how far it can be pushed safely

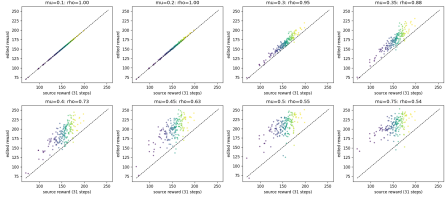
Statistics ($n = 400$ windows):

$$y_i = \beta_0 + \beta_Q(Q_i - \tfrac{1}{2}) + \beta_S(S_i - \tfrac{1}{2}) + \beta_{QS}(Q_i - \tfrac{1}{2})(S_i - \tfrac{1}{2}) + e_i$$

95% CI: 2,000 bootstrap resamples; p: 2,000 permutations.

Decision rule (fixed in advance): **supported** if CI(β_S)> 0, CI(β_Q)< 0 and the extreme-quadrant gap ≥ 0.1.

Pilot (S_1, 184 windows): ceiling effect hid support



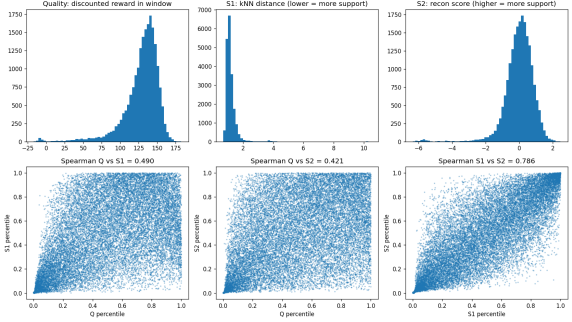
At $\mu \geq 0.45$ edits converge to ≈ 175–250 reward whatever the source.

μ^* analysed with	β_Q [95% CI]	β_S [95% CI]
all μ, $\theta = 0.017$	-0.17 $[-0.24, -0.10]$	$+0.02$ $[-0.05, 0.09]$
all μ, $\theta = 0.0085$	-0.15 $[-0.22, -0.08]$	-0.01 $[-0.08, 0.07]$
$\mu \leq 0.5$, $\theta = 0.017$	-0.02 $[-0.05, 0.00]$	$+0.01$ $[-0.02, 0.05]$
$\mu \leq 0.5$, $\theta = 0.0085$	-0.01 $[-0.05, 0.02]$	$+\mathbf{0.09}$ $[0.04, 0.14]$

- The apparent quality effect came from $\mu = 0.75$ edits that rewrite everything.
- Restricting to $\mu \leq 0.5$ reveals a support effect.
- **Fixed for main run:** $\mu \leq 0.5$, $\theta = 0.0085$, support S_2.

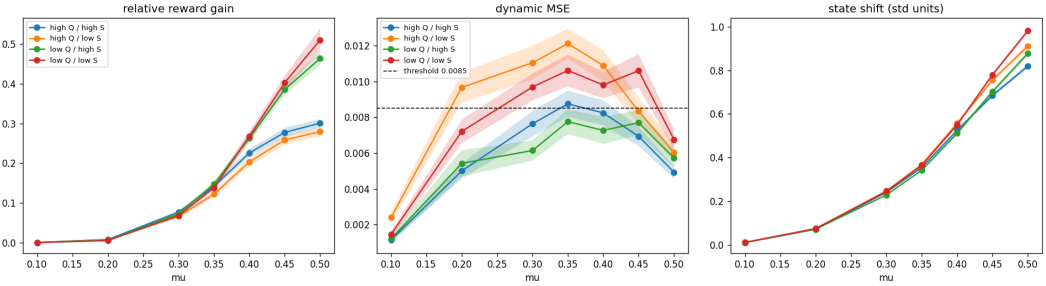
Main study (S_2): data funnel

stage	count	how
transitions	1,000,000	1,000 episodes $\times$ 1,000
eligible windows (start ≤ 800)	801,000	RTG not truncated
pool	20,000	random; Q, S_1, S_2 computed (S_3 on 100)
selected windows	**400**	5×5 grid on $Q \times S_2$ quintiles, 16 per cell
edits analysed	**5,600**	400×7 μ (≤ 0.5) $\times$ 2 draws
simulated transitions	198,400	every edit replayed in MuJoCo



GTA $\rightarrow$ per-window μ, α $\rightarrow$ results so far

Main result: gain follows quality, physics error follows support



- **Reward gain** rises with μ for all; low-Q windows gain most (ceiling effect).
- **Physics error** is higher for *low-support* windows at every μ, peaking at $\mu \approx 0.35$.
- **Shift** is the same for all quadrants: equal edit size, unequal damage.

Correlations at every μ (Spearman, 400 windows)

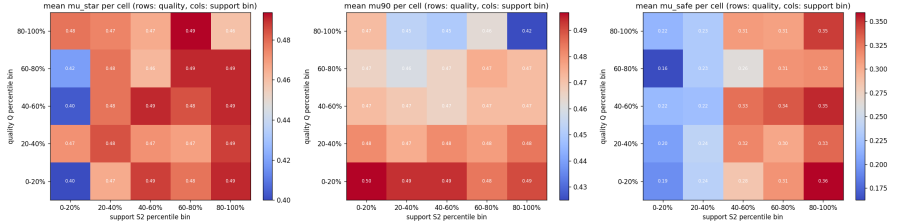
μ	0.1	0.2	0.3	0.35	0.4	0.45	0.5
S_2 vs physics error	-0.29	-0.36	-0.34	-0.28	-0.24	-0.18	-0.13
S_1 vs physics error	-0.25	-0.34	-0.32	-0.26	-0.28	-0.25	-0.15
Q vs physics error	$+0.07$	$+0.01$	$+0.01$	$+0.09$	$+0.04$	-0.14	-0.07
S_2 vs reward gain	$+0.11$	$+0.11$	$+0.12$	$+0.14$	$+0.09$	$+0.06$	$+0.06$
Q vs reward gain	$+0.13$	$+0.17$	$+0.01$	-0.11	-0.25	-0.51	-0.65
mean reward gain	0.000	0.007	0.071	0.138	0.240	0.331	0.388
mean physics error	0.0015	0.0068	0.0086	0.0098	0.0090	0.0084	0.0058
share over θ	0.02	0.24	0.36	0.43	0.41	0.33	0.18

Both support metrics predict physics error at every μ; quality does not. Note the support effect **fades towards** $\mu = 0.5$: at high noise the model regenerates the window and forgets the source.

Test of H2 at window level

outcome	β_S [95% CI]	p_S	β_Q [95% CI]	gap	verdict
μ^*	$+0.059$ $[0.030, 0.090]$	< 0.001	-0.001 $[-0.025, 0.022]$	$+0.034$	S only, small
μ_{90}	-0.002 $[-0.014, 0.009]$	0.78	-0.048 $[-0.063, -0.034]$	$+0.020$	Q only, small
μ^{safe} †	$+\mathbf{0.183}$ $[0.139, 0.225]$	< 0.001	-0.013 $[-0.061, 0.036]$	$+\mathbf{0.109}$	S only

† exploratory (defined after first look). μ^* is pinned at the grid top: 279 of 400 windows have $\mu^* = 0.5$.



μ^{safe} by support quintile: **0.20, 0.23, 0.30, 0.31, 0.34** — in every quality row. This curve is what we used to assign μ next.

Experiment A: assigning μ by support quintile

Source windows	**161,291** random windows ($\times 31 \approx 5$M generated transitions). **Same windows in every condition**; only μ differs.
Support	S_2 reconstruction score, quintiles within these windows
RL	TD3+BC (CORL), batch 1024, 1M steps, 3 seeds; score = mean of last 5 evals

condition	μ per support quintile (low $\rightarrow$ high)					mean
`none`	original data only					–
`uniform_0.3`	0.3	0.3	0.3	0.3	0.3	0.3
`adaptive`	0.20	0.25	0.30	0.35	0.40	0.3
`inverted`	0.40	0.35	0.30	0.25	0.20	0.3
`uniform_0.5`	0.5	0.5	0.5	0.5	0.5	0.5
`adaptive_wide`	0.3	0.4	0.5	0.6	0.7	0.5
`inverted_wide`	0.7	0.6	0.5	0.4	0.3	0.5

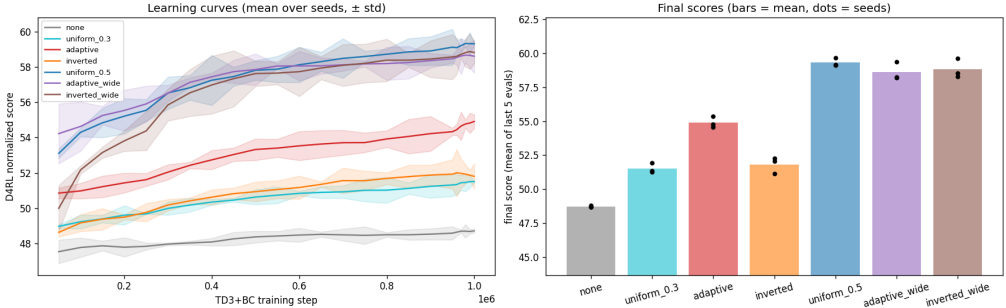
`adaptive` follows the μ^{safe} curve. `inverted` uses the same values in the wrong order: controls for mixing μ values.
Rule: **supported** if adaptive beats *both* same-mean controls (Welch, $p < 0.05$).

Experiment A results: two TD3+BC implementations

condition	PyTorch (CORL)	JAX port	data: dyn. MSE / gain
`none`	48.65 ± 0.18	48.72 ± 0.06	–
`uniform_0.3`	51.02 ± 0.66	51.51 ± 0.36	0.0096 / 0.057
`adaptive`	$\mathbf{53.53 \pm 0.37}$	$\mathbf{54.91 \pm 0.41}$	0.0090 / 0.079
`inverted`	51.72 ± 0.49	51.81 ± 0.60	0.0091 / 0.083
`uniform_0.5`	$\mathbf{58.67 \pm 0.45}$	$\mathbf{59.31 \pm 0.30}$	0.0076 / 0.327
`adaptive_wide`	–	58.60 ± 0.65	0.0092 / 0.220
`inverted_wide`	–	58.81 ± 0.70	0.0075 / 0.319

comparison	PyTorch	JAX
adaptive − uniform_0.3	$+2.51$ ($p = 0.009$)	$+3.40$ ($p < 0.001$)
adaptive − inverted	$+1.81$ ($p = 0.009$)	$+3.10$ ($p = 0.003$)
adaptive − uniform_0.5	-5.14 ($p < 0.001$)	-4.40 ($p < 0.001$)
adaptive_wide − uniform_0.5	–	-0.71 ($p = 0.20$)
adaptive_wide − inverted_wide	–	-0.21 ($p = 0.72$)

Experiment A: learning curves (JAX)



- At mean $\mu = 0.3$: support-adaptive is clearly better than both controls, **replicated** across two implementations (H2c **supported**).
- At mean $\mu = 0.5$: no benefit. Pushing high-support windows to $\mu = 0.7$ *raises* physics error (0.0092 vs 0.0076) and lowers gain.
- Consistent with the window study: the support effect is strongest at $\mu \approx 0.2$–0.35 and vanishes near 0.5.

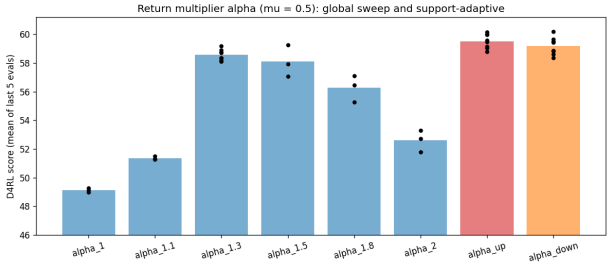
From μ to α: design

Since μ-adaptation did not help at $\mu = 0.5$, we fixed $\mu = 0.5$ and varied the **return multiplier** instead.

Data	32,258 windows (subset of the 161k, seed 2) $\Rightarrow$ 1M generated transitions; JAX TD3+BC, 1M steps
Stage 1	global $\alpha \in \{1.0, 1.1, 1.3, 1.5, 1.8, 2.0\}$, 3 seeds $\Rightarrow$ best α^*
Stage 2	α by S_2 quintile around α^*: `alpha_up` $= 1.1, 1.2, 1.3, 1.4, 1.5$ (low$\rightarrow$high support), `alpha_down` reversed; 8 seeds
Stage 3	control: `alpha_random` uses the same five values assigned *at random*; fresh seeds 10–14

Reading: if `alpha_up` $>$ `alpha_random`, support matters. If `alpha_random` $\approx$ `alpha_up` $>$ uniform, only *mixing* α values matters.

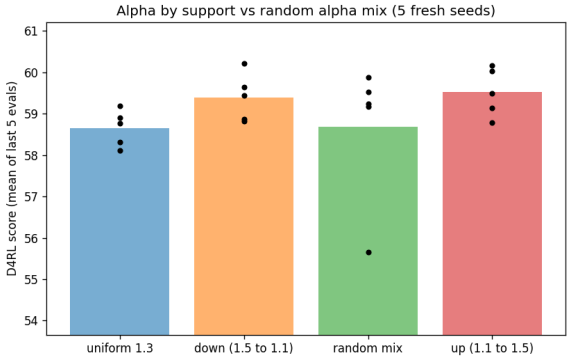
α results: Stage 1 and 2



condition	score	seeds
$\alpha = 1.0$	49.14 ± 0.14	3
$\alpha = 1.1$	51.35 ± 0.14	3
$\alpha = 1.3$	$\mathbf{58.57 \pm 0.38}$	8
$\alpha = 1.5$	58.10 ± 1.10	3
$\alpha = 1.8$	56.28 ± 0.93	3
$\alpha = 2.0$	52.61 ± 0.77	3
`alpha_up`	$\mathbf{59.53 \pm 0.51}$	8
`alpha_down`	59.18 ± 0.62	8

- Interior optimum at $\alpha = 1.3$ (matches the paper). $\alpha = 1$ ($\approx$ no guidance push) is barely better than no augmentation.
- up $-$ 1.3: $+0.96$ ($p = 0.001$); down $-$ 1.3: $+0.61$ ($p = 0.035$); up $-$ down: $+0.35$ ($p = 0.24$).
- Both directions help $\Rightarrow$ suspect it is the *mix*, not the support ordering.

α results: Stage 3 (random-mix control)



condition (5 fresh seeds)	score
$\alpha = 1.3$	58.66 ± 0.44
`alpha_down`	59.40 ± 0.58
`alpha_random`	58.69 ± 1.72
`alpha_up`	59.52 ± 0.58

- up $-$ random: $+0.83$ ($p = 0.35$)
- up $-$ down: $+0.12$ ($p = 0.74$)
- up $-$ 1.3: $+0.87$ ($p = 0.031$)
- random $-$ 1.3: $+0.04$ ($p = 0.97$)

Reading: `alpha_up` reliably beats uniform $\alpha = 1.3$ (replicated on fresh seeds), but cannot be separated from the random mix (high variance, 1.72) or from `alpha_down`. Evidence that the support *ordering* matters for α: **inconclusive**.

Learning continuous $\mu(S, Q)$ and $\alpha(S, Q)$

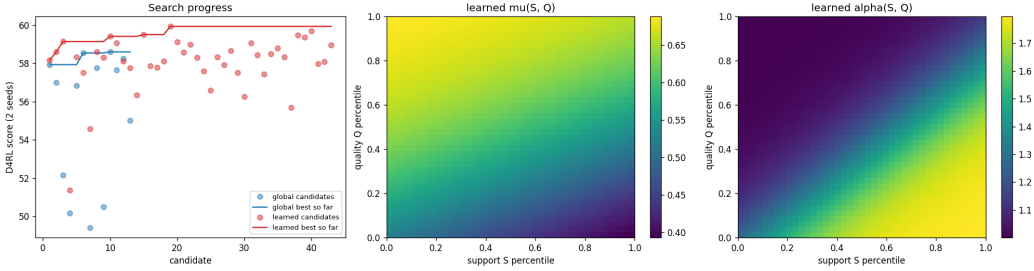
Instead of hand-picked quintile maps, a smooth map with 6 parameters:

$$\mu_i = 0.3 + 0.4\,\sigma(a_0 + a_S S'_i + a_Q Q'_i), \qquad \alpha_i = 1.0 + 0.8\,\sigma(c_0 + c_S S'_i + c_Q Q'_i)$$

$S', Q' \in [-\frac{1}{2}, \frac{1}{2}]$ centred percentiles; $\mu \in [0.3, 0.7]$, $\alpha \in [1.0, 1.8]$. Per-window μ implemented by noising each window at its own $\sigma_{k_0(i)}$ inside one batched sampler.

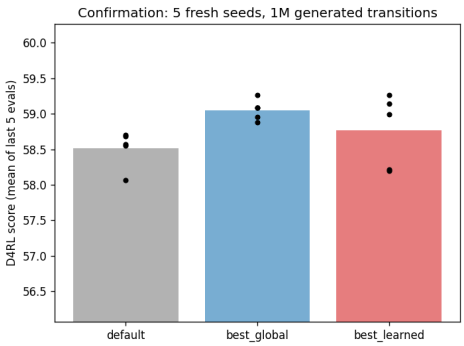
Optimiser	CMA-ES on the D4RL score (black box)
global search	only $[a_0, c_0]$ (best single μ, α): 13 candidates
learned search	all 6 parameters: 43 candidates
Search budget	16,129 windows (500k transitions), 2 seeds; common random numbers across candidates
Confirmation	1M transitions, 5 **fresh** seeds (20–24); rule: learned must beat best global ($p < 0.05$)

Learned map: search and what it found



- Best learned (search): 59.93 vs 58.60 for best global — but on only 2 seeds and selected from 43 candidates.
- Learned μ: *decreases* with support, increases with quality ($a_S = -1.4$, $a_Q = +3.3$) — opposite of H2a/H2b.
- Learned α: strongly up with support, down with quality ($c_S = +6.2$, $c_Q = -7.0$): push hard only well-supported, low-quality windows.

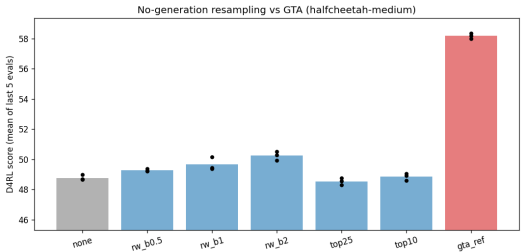
Learned map: confirmation on fresh seeds



setting	score (5 seeds)	mean μ	mean α
default	58.51 ± 0.26	0.500	1.300
best global	$\mathbf{59.05 \pm 0.15}$	0.565	1.356
best learned	58.76 ± 0.52	0.592	1.277

- learned − global: -0.29 $(p = 0.29) \Rightarrow$ **not supported**.
- global − default: $+0.54$ $(p = 0.006)$: a slightly larger single μ, α helps.
- Search advantage was a winner's curse (2 seeds, 43 tries).
- Learned data has higher physics error (0.0091 vs 0.0066).

H3: is the diffusion model needed at all?



5M extra transitions *resampled* from the original data (no generation):

condition	score	% of GTA gain
none	48.77	0
reward-weighted $\beta = 0.5$	49.27	5
reward-weighted $\beta = 1$	49.66	9
reward-weighted $\beta = 2$	50.24	16
top-25% windows	48.53	−3
top-10% windows	48.85	1
GTA (μ 0.5, α 1.3)	**58.18**	100

Reweighting existing data recovers at most 16% of GTA's gain: the gain comes from **new, higher-reward** states, not from re-emphasising good ones.

H4/H5: can a cheaper editor (VAE) replace diffusion?

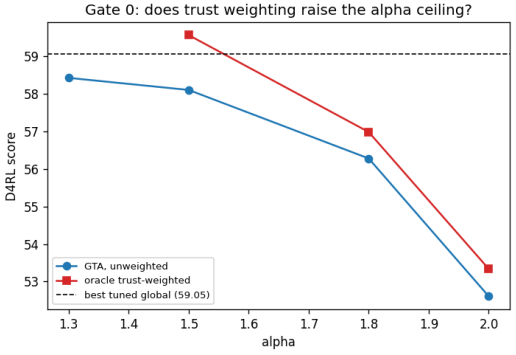
H4: return-conditioned VAE edits windows in latent space (1,000-window simulator check; GTA ref: $+32\%$, dyn. MSE 0.0075).

method	gain	dyn. MSE	shift	TD3+BC
VAE reconstruction	$+0.5\%$	0.032	0.21	–
VAE edit $\alpha = 1.3$, $\lambda = 1$	$+27.8\%$	0.046	0.39	48.51 (3 seeds)
VAE edit $\alpha = 1.5$, $\lambda = 1$	$+43.9\%$	0.087	0.61	–
GTA $\mu = 0.5$, $\alpha = 1.3$	$+32\%$	**0.0075**	0.89	**58.5**

H5: keep VAE-edited actions, roll out states with a learned dynamics ensemble: physics is consistent but reward **drops** (-10 to -26%).

Reading: the VAE gains reward but with $6\times$ the physics error, and TD3+BC gets nothing from it (48.5 = no augmentation). Physical validity matters more than reward gain.

H6: trust-weighting generated transitions (Gate 0, oracle)

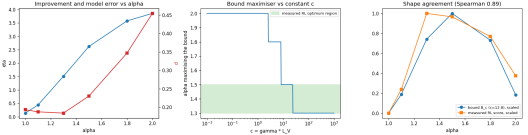


Weight per generated transition $w = \exp(-\text{err}/0.0085)$ with the *true* physics error (oracle); original data weight 1.

α	unweighted	weighted	p
1.5	58.10	**59.56**	0.13
1.8	56.28	56.98	0.32
2.0	52.61	53.34	0.29

Gate passed: best weighted score 59.56 exceeds best tuned global (59.05). Differences individually not significant (3 seeds). Next: replace the oracle with a learned error estimate (e.g. the S_2 score of the *edited* window).

H7: does a simple bound explain the α optimum?



$$B_c(\alpha) = \frac{\eta(\alpha) - c\, d(\alpha)}{1 - \gamma}$$

η: true reward gain per step; d: next-state error; $c = \gamma L_V$.

α	η	d	RL score
1.0	0.136	0.193	49.14
1.3	1.511	0.184	**58.42**
1.5	2.621	0.230	58.10
2.0	3.858	0.455	52.61

Any $c \in [8.3, 10^3]$ (incl. the worst-case 824) puts the maximiser at $\alpha = 1.3$–1.5; best rank fit $\rho = 0.89$. Explanatory only (c fitted on one dataset): α trades reward gain against physics error, the same trade-off support measures.

Scorecard

claim	evidence	result
Reconstruction score S_2 measures support	$\rho = 0.79$ with kNN, 0.74 with exact likelihood	**supported**
H2a support $\Rightarrow$ safe μ	μ^{safe} $+0.18$ $[0.14, 0.23]$; physics error vs S at every μ	**supported**
H2b quality $\Rightarrow$ lower μ	$\beta_Q \approx 0$; quality only drives gain (ceiling)	**not supported**
H2c adaptive μ, mean 0.3	$+2.5/+3.4$ over controls, 2 implementations	**supported**
H2c adaptive μ, mean 0.5	-0.7 ($p = 0.20$)	**not supported**
α by support	beats uniform 1.3 ($+0.9$) but $\approx$ random mix	inconclusive
Learned $\mu(S, Q), \alpha(S, Q)$	-0.3 vs tuned global ($p = 0.29$)	**not supported**
H3 generation needed	resampling recovers $\leq 16\%$	**supported**
H4/H5 VAE editor	no RL gain; $6\times$ physics error	**not supported**
H6 trust weighting (oracle)	59.56 vs 59.05	gate passed

Takeaway: support predicts edit damage; it pays off for small μ. Best score so far: **59.5** (α mix / oracle weighting) vs 58.5 GTA default.

Limitations and next steps

Limitations

- One dataset (`halfcheetah-medium-v2`), one behaviour policy: support varies only moderately.
- 3–5 seeds: differences below ≈ 1 point are hard to resolve (e.g. random α mix has std 1.7).
- Guidance at $w = 0$ instead of the paper's $w = 2$; condition dropout active during sampling (upstream).
- μ^{safe} and the wide/α designs were defined after earlier results (exploratory).

Next steps

- Mixed-quality datasets (medium-replay, medium-expert) where support varies much more.
- H6 with a learned, non-oracle trust weight (post-edit S_2 or dynamics-ensemble disagreement).
- Quality-indexed μ (`quality_adaptive`/`quality_inverted`, already implemented, not yet run).
- More seeds for the α up/random/down comparison.

Reproducibility

command	produces
`python scripts/h2_run.py pool | sweep | analyze`	window-level study (`outputs/h2/`)
`bash scripts/h2_rl_all.sh`	Experiment A, PyTorch (`outputs/h2_rl/`)
`bash scripts/h2_rl_jax_all.sh`, `h2_rl_wide.sh`	Experiment A, JAX + wide (`outputs/h2_rl_jax/`)
`bash scripts/h2_alpha.sh`, `h2_alpha_stage3.sh`	α stages 1–3 (`outputs/h2_alpha/`)
`python scripts/h2_learn.py run`	CMA-ES learned map (`outputs/h2_learn/`)
`bash scripts/h3_nogen.sh`, `h4_vae_rl.sh`, `h6_gate0.sh`	H3, H4, H6
`python scripts/h5_vae_dyn.py`, `h7_t1.py`	H5, H7

Checkpoint `model-1000000.pt` (EMA). Code: `src/h2/` (`support`, `analysis`, `rl_data`, `learned`, ...), `jax_impl/`.